

Interface lattice displacement measurement to 1 pm by geometric phase analysis on aberration-corrected HAADF STEM images

Yuanyuan Zhu^a, Colin Ophus^b, Jim Ciston^b, Haiyan Wang^{a,c,*}

^a Program of Materials Science and Engineering, Texas A&M University, College Station, TX 77843-3128, USA

^b National Center for Electron Microscopy, Lawrence Berkeley National Laboratory, Berkeley, CA 94720, USA

^c Department of Electrical and Computer Engineering, Texas A&M University, College Station, TX 7843-3128, USA

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Abstract

In this work, the accuracy of geometric phase analysis (GPA) on aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (Cs-corrected HAADF-STEM) images for lattice strain measurement at heterogeneous interfaces has been systematically investigated. Starting with an ideal crystal lattice of synthetic images, and then experimental HAADF images of a single-crystal lattice, we have quantitatively evaluated the inherent GPA processing artifacts and experimental errors due to STEM scanning distortions. Our results suggest that, with a properly chosen Fourier mask size and strain profile direction/width, 1 pm accuracy can be achieved for GPA strain quantification in the STEM fast-scan direction with a spatial resolution of <1 nm. To demonstrate the effectiveness and reliability of the STEM-based GPA strain profile, we have applied it to two experimental heterointerfaces: the strained LaAlO₃/SrTiO₃ (LAO/STO) and the relaxed SrTiO₃/MgO (STO/MgO). Interestingly, GPA strain mapping reveals a novel secondary relaxation mechanism in the LAO/STO heterostructures. Essential limitations in GPA are also discussed using the example of a FeSe_{0.5}Te_{0.5}(FST)/SrTiO₃ heterointerface. Although we focus on the interfacial lattice strain in this paper, the approaches for strain error estimation and the fundamental discussions on line profiles can also be applied, with some modifications, to other nanostructures.

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1. Introduction

The effects of lattice strain fields on the nature of heterogeneous interfaces have been a long-standing question in studies of the mechanical and physical properties of thin films [1,2]. The film strain effect has attracted extensive research interest owing to the recent discovery of exotic transport properties, e.g. insulator–metal transitions [3,4], abnormal magnetoresistance [5] and superconductivity [6] in functional heterogeneous structures. For example, investigations of prototype perovskite SrTiO₃ (STO) showed

that lattice distortion in favor of polarization-related tetragonality could enable room-temperature ferroelectricity [7]; studies on novel iron-based superconducting Fe_{1+y}Te_{1-x}Se_x thin films found that interface strain modification [8] can either boost the critical superconducting transition temperature [9,10] or destroy the superconductivity [11]. Control of the interface lattice mismatch strain requires an accurate knowledge of the actual strain field of the heterostructures on the atomic scale.

Recent progress in digital image processing has enabled routine extraction and quantification of lattice parameters at an extremely high spatial resolution using dot-like high-resolution images. Geometric phase analysis (GPA) [12] and peak finding [13,14] are the two main methods, working from Fourier space and real space, respectively. Despite their algorithmic differences, the fundamental idea of

* Corresponding author at: Program of Materials Science and Engineering, Texas A&M University, College Station, TX 77843-3128, USA. Tel.: +1 9798455082; fax: +1 9798456259.

E-mail address: wangh@ece.tamu.edu (H. Wang).

image-based strain measurements is to identify the image contrast maxima as a periodic lattice and to calculate (and illustrate) the deviation of a real local lattice with respect to a reference crystal. These methods rely on the assumption that the image intensity peaks directly correspond to the positions of atomic columns in a given projection [14]. However, this assumption is not always valid, especially for phase-contrast-dominant high-resolution transmission electron microscopy (HRTEM) which is known for its contrast reversal with sample thickness and microscope focus [15,16]. In this work, high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) has been chosen over HRTEM images in determining lattice strain at defective heterogeneous interfaces for the following reasons: (i) the contrast of STEM images is less sensitive to local sample variations (see Figs. S1 and S2 for a comparison of HRTEM and HAADF-STEM images of the same heterointerface sample)—it has been experimentally confirmed that no contrast reversal occurs in STEM for a wide range of focus and foil thickness [17]; (ii) STEM allows high resolution images to be taken from thicker foils to minimize surface relaxation [18,19]; (iii) Z-contrast STEM images add another dimension for mapping lattice chemistry distribution at the same time.

The goal of this work is to experimentally assess the accuracy of GPA for the measurement of lattice strain at thin-film heterogeneous interfaces using Cs-corrected HAADF-STEM images and to provide practical guidelines for strain profiling. A systematic evaluation has been conducted, starting with computer-generated images with ideally sharp contrast, free of any distortion (the perfect crystal lattice), which were used to explore the inherent inaccuracy of GPA digital processing. Next, GPA was applied on the Cs-corrected STEM images of defect-free single-crystalline lattices (the experimental crystal lattice) to determine the error bar of GPA strain measurement introduced by STEM imaging. Scanning distortions in the slow-scan (the “fly-back” error [20,21]) and in the fast-scan directions were examined with a complementary evaluation by the real space method. Then, two case studies, i.e. GPA strain mapping on Cs-corrected STEM images of two representative heterogeneous interfaces, the strained $\text{LaAlO}_3/\text{SrTiO}_3$ (LAO/STO) interface and the relaxed $\text{SrTiO}_3/\text{MgO}$ (STO/MgO) interface, were presented. Finally, limitations with GPA strain quantification are discussed in terms of the crystal structures studied.

2. Materials and methods

2.1. Synthetic images generated by computer

Artificial images of a two-dimensional dot-like lattice were generated using Origin Pro 8.5 (similar to Ref. [22]). To match these synthetic images with the experimental thin-film samples, lattice symmetries of perovskite SrTiO_3 and rocksalt MgO on the [100] zone axis were applied.

As shown in Fig. 1a (and enlarged in Fig. 1b), combining two such dot-contrast images with different lattice spacings results in a heterogeneous interface with rigid contact with well-defined in-plane x - and out-of-plane y -coordinates. Computed image #1 was designed as a coherent interface (within the field of view), with small lattice differences between the substrate (23.50 pixels per unit cell) and the film (23.33 pixels per unit cell), as illustrated in the line profile across the interface in Fig. 1c. Images of an incoherent interface with different misfit dislocation densities, i.e. computed images #2 and #3 (a model of the STO/MgO interface) were obtained by modifying the film lattice spacing with the respect to that of the substrate. Note that no lattice strain is being introduced on either side of the interface. All three computed images are 1024×1024 pixels in size.

2.2. Thin-film and TEM sample preparation

Thin-film samples were grown by a standard pulsed-laser deposition technique (with a KrF excimer laser, $\lambda = 248$ nm) using commercial LAO and STO targets (CERAC, Inc.), a homemade $\text{FeSe}_{0.5}\text{Te}_{0.5}$ (nominal) target, and commercial single-crystal STO(100) and MgO(100) substrates. The depositions were carried out at the same substrate-to-target distance of 5 cm, target temperature of 780°C , laser energy of 300 mJ, repetition rate of 2 Hz and oxygen pressure of 27 Pa for the LAO and STO thin films. After deposition for 600 and 720 s, the LAO/STO and STO/MgO films were directly cooled down to room temperature in an atmosphere of 4×10^4 Pa (300 torr) O_2 without post-annealing. The $\text{FeSe}_{0.5}\text{Te}_{0.5}$ film was grown on single-crystal STO(100) substrates in a controlled oxygen atmosphere ($\sim 10^{-2}$ Pa; details on target processing and deposition parameters can be found elsewhere [23]). The overall LAO, STO and $\text{FeSe}_{0.5}\text{Te}_{0.5}$ film thicknesses are around 50, 40 and 150 nm, respectively. The overall microstructure properties of the LAO and STO films were confirmed by X-ray diffraction (XRD; Bruker D8 powder X-ray diffractometer). The $\text{FeSe}_{0.5}\text{Te}_{0.5}$ film was examined by selected-area electron diffraction (SAED) using a JEOL JEM-2010 microscope. Cross-sectional specimens of the heterointerfaces were prepared using a conventional TEM sample preparation routine along the $\langle 100 \rangle$ direction. After cutting and gluing, the TEM specimens were then ground, plane-parallel polished and further thinned in the center by dimpling. Ar ion milling was used to obtain a perforation and electron-transparent thin area with a Gatan Precision Ion Polishing System (Gatan, Pleasanton, CA). All TEM samples were plasma cleaned each time with a 25% O_2 and 75% Ar mixture before insertion into the microscope.

2.3. Cs-corrected STEM experimental setup

In this work, TEAM 0.5 [24], a modified FEI Titan microscope with a special high-brightness Schottky field-

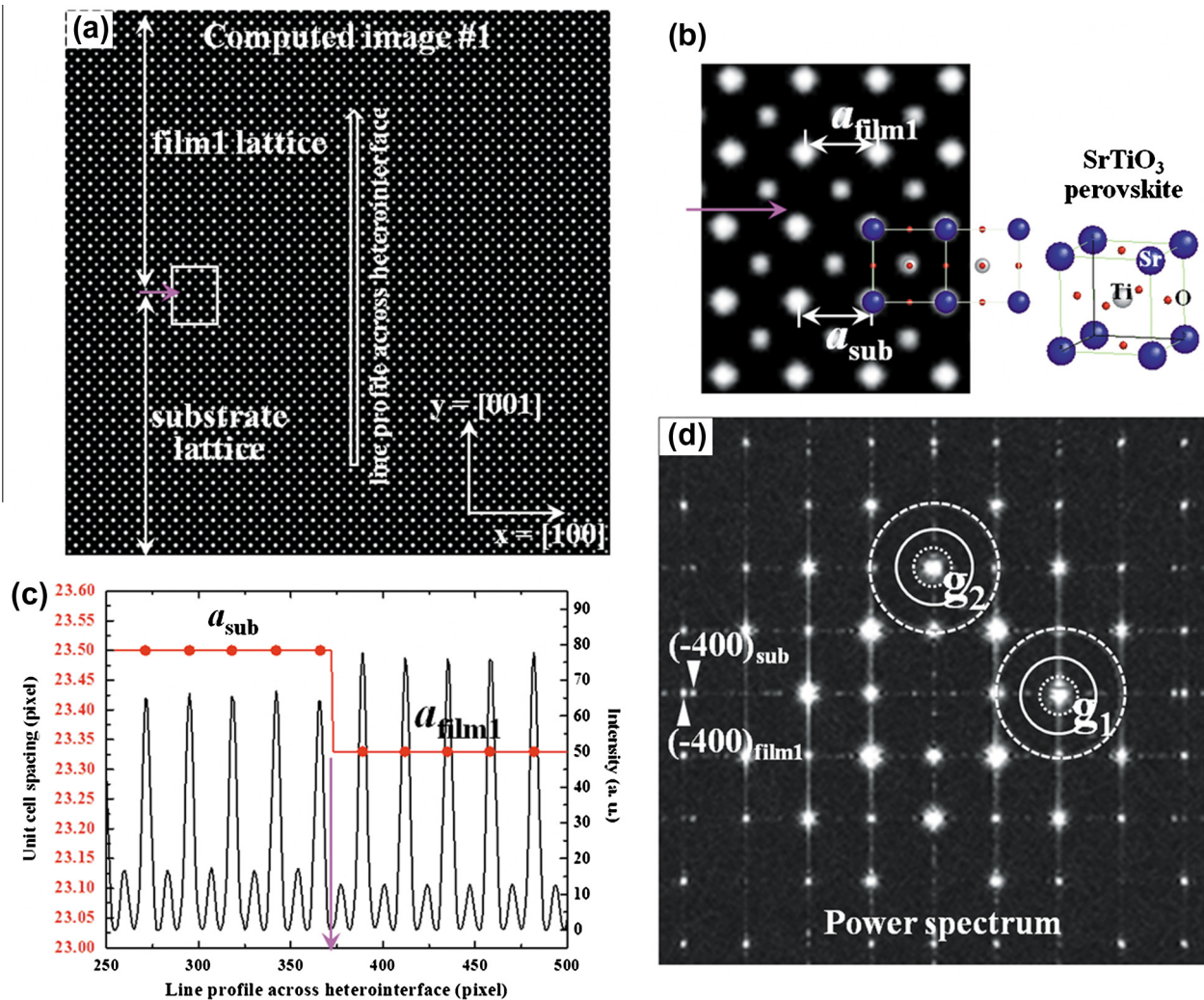


Fig. 1. (a) Computed image #1, a computer-generated 2-D lattice of a heterogeneous structure with a small lattice mismatch. The image is in (100). (b) A magnified image of the rigid interface between film #1 and the substrate, both of which are in perovskite SrTiO₃ structure with contrast maxima at atomic positions. (c) Unit-cell spacing and intensity line profile along the y-direction across the interface. (d) The power spectrum of (a). The Fourier vectors selected for GPA are noted as g_1 and g_2 . Three Fourier masks with different sizes are marked by circles.

emission electron source and an improved hexapole-type illumination aberration corrector, was employed. By correcting the spherical aberration up to the fifth order using the illumination aberration corrector, TEAM 0.5 allows a larger beam convergence angle which provides a small electron probe of high current. One of main advantages of the aberration-corrected STEM is thus a significant improvement in the spatial resolution to as much as 0.05 nm (which is also the electron probe size) [25]. This sub-ångström electron probe is much smaller than a typical lattice spacing of 0.2 nm, which helps minimize the transverse interference effects between neighboring atomic columns [26]. All HAADF STEM micrographs in this paper were recorded using the TEAM 0.5 with a convergence semi-angle of 17 mrad after fine-tuning of the probe corrector at 300 kV to a flat-phase angle of over 25 mrad (values for typical residual hardware aberrations are given in [27]). Unless stated otherwise, an ADF detector inner semi-angle of 68 mrad was used for the Z-contrast imaging, which

gave reasonable signal-to-noise ratios. The microscope is equipped with a bottom-mounted 2048 × 2048 pixel slow-scan CCD camera; all the HAADF images were recorded at a 1024 × 1024 pixel image size, using a dwell time of 10 μs (unless otherwise specified).

TEM samples were left in the microscope for a sufficiently long time to reduce thermal drift (usually 1800–3600 s until no noticeable drifting was observed). The local TEM sample thickness was estimated by low-loss electron energy loss (EELS) spectra based on a calculation of the inelastic mean-free path derived from the effective sample atomic number [28].

2.4. Geometric phase analysis and real space lattice measurement

In this work, a commercial program, a Digital Micrograph plug-in (DM 1.8.3 package, HREM Research Inc.), was used for GPA. Developed by Hÿtch and

colleagues [12], the geometric phase algorithm reconstructs the displacement field utilizing Fourier filtering centered around two non-collinear Bragg vectors of the power spectrum generated from a high-resolution micrograph. The vectorial representation of the displacement field is:

$$u(r) = -(1/2\pi)[Pg_1(r)a_1 + Pg_2(r)a_2], \quad (1)$$

where r is a position in the image, and a_1 and a_2 are the associated lattice vectors of g_1 and g_2 in real space. $Pg_1(r)$ and $Pg_2(r)$ are two geometric phase images representing the local deviation in two sets of lattice planes selected from a reference lattice. The “phase” in GPA does not refer to the electron wave function, but to the position of image contrast maxima; hence it is called the “geometric” phase. In this study, for the cubic structures two low-order vertical vectors $g_1 = [020]^*$ and $g_2 = [002]^*$ were usually selected. Then, the derivation of the displacement field gives the strain field in a principal direction:

$$\begin{aligned} \varepsilon_{xx} &= \partial u_x(r)/\partial x, \quad \varepsilon_{yy} = \partial u_y(r)/\partial y \text{ and} \\ \varepsilon_{xy} &= 1/2(\partial u_x(r)/\partial y + \partial u_y(r)/\partial x) \end{aligned} \quad (2)$$

Thus the biaxial strains ε_{xx} (in-plane), ε_{yy} (out-of-plane) and ε_{xy} (shear strain) are derived to illustrate the local lattice displacement from the reference lattice. For additional mathematical and theoretical discussion of GPA, see Refs. [12,29].

Note that the strain determined by GPA here is a relative value. For example, in the case of interface lattice strain, where the substrate lattices are usually selected as reference, $\varepsilon_{xx}^{\text{GPA}} = \varepsilon_{xx}^{\text{relative}} = (a^{\text{local}} - a^{\text{sub}})/a^{\text{sub}}$; the sign of ε indicates whether the measured local lattice is larger (“+”) or smaller (“−”) than the reference lattice. In this study, reference regions with a size of 256×256 pixel were selected for all 1024×1024 pixel images (and 128×128 pixel reference for the images 512×512 pixels in size). The relative strain can be easily transformed into the conventional strain when the reference lattice parameter and the bulk material lattice are known, $\varepsilon_{xx}^{\text{absolute}} = (a^{\text{local}} - a^{\text{bulk}})/a^{\text{bulk}}$; or one can simply calculate the actual local lattice parameters (of the thin film, as the substrates are usually a standard crystal with a known lattice parameter) for further analysis. GPA generates strain maps with color contours directly illustrating the location of the relative strain. In this paper, a scale range of -10% to $+10\%$ was applied to all strain maps for consistency.

A real space strain method, developed in house, was also applied to the experimental crystal lattice to determine the error bar of the strain measurement as a function of kernel size (and STEM scan direction and dwell time). This method consisted of three main steps: (i) sub-pixel peak fitting via 2-D Gaussian functions; (ii) lattice fitting via linear algebra; (iii) conversion of discrete measurements into a 2-D displacement image using kernel density estimation. In the third step, a Gaussian kernel was adopted since it can be smoothly differentiated.

3. Results and discussion

3.1. GPA of computed images

We first investigated the factors contributing to the inherent inaccuracy of the digital processing GPA strain quantification using computed images. Starting with the simplest scenario of a coherent heterointerface in Fig. 1a, the influences of one of the most important GPA processing parameters, the Fourier mask size, on strain mapping were studied. As shown in the power spectrum of computed image #1 in Fig. 1d, we tested the largest mask size possible, i.e. $1/a_{\text{sub}}$, a small mask of $1/4a_{\text{sub}}$, and an intermediate mask size, $1/2a_{\text{sub}}$. The GPA strain maps of ε_{xx} (in-plane), ε_{yy} (out-of-plane) and ε_{xy} (shear strain) produced by these three masks using the substrate lattice as reference are presented in Fig. 2a–i. Despite all images being generated from the same source image of Fig. 1a, the strain maps become increasingly rougher, with increasing noise as the GPA mask size increases. For comparison, strain profiles across the interface with a fixed width of 600 pixels were applied (see the example in Fig. 2d) and are plotted in Fig. 3a–c.

Prior to discussing the mask size effect, it should be noted that the profile direction and position can also affect the strain value extracted. Within a set of three strain profiles, e.g. the medium mask in Fig. 2j, ε_{yy} presents more severe oscillations than ε_{xx} and ε_{xy} even away from the interface. This is because the strain variations in ε_{xx} are along the direction parallel to the interface, but ε_{yy} oscillations are mainly perpendicular to the interface. When applying a line profile across the interface, the ε_{xx} oscillations cancel out among the 600-pixel sampling, while the variations in ε_{yy} are preserved. If the line profile is carried out along the interface, the results will be reversed. For the shear strain ε_{xy} , although its variations have the same direction as ε_{yy} , they are essentially very small (in a range of $\pm 0.01\%$ in Fig. 3c), and therefore the profile direction effect is not obvious. Since the lattice strain of the heterogeneous interface is the main topic of this study, there are some “default” settings, e.g. the profile direction is always across the interface, in order to compare the substrate and the film, similar to several previous studies [30–33]. Also, these ways of sampling can present some differences—for example, the ε_{yy} profile often seems noisier than that of the ε_{xx} profile. These phenomena can vary if one applies the strain profile in a different way on lattice images of other structures such as nanoparticles. However, it is always important to know whether the strain variations are by-products of a quantifying process or representative of the imaged lattice.

To clarify the influence of the mask size on the strain profile, the substrate regions (marked by rectangles) in Fig. 3a–c were enlarged and are presented in Fig. 3d–f, in which the features of the strain profiles are directly related to the selection of mask size. In Fig. 3d, all three in-plane strain ε_{xx} profiles exhibit sinusoidal-wave-like oscillations

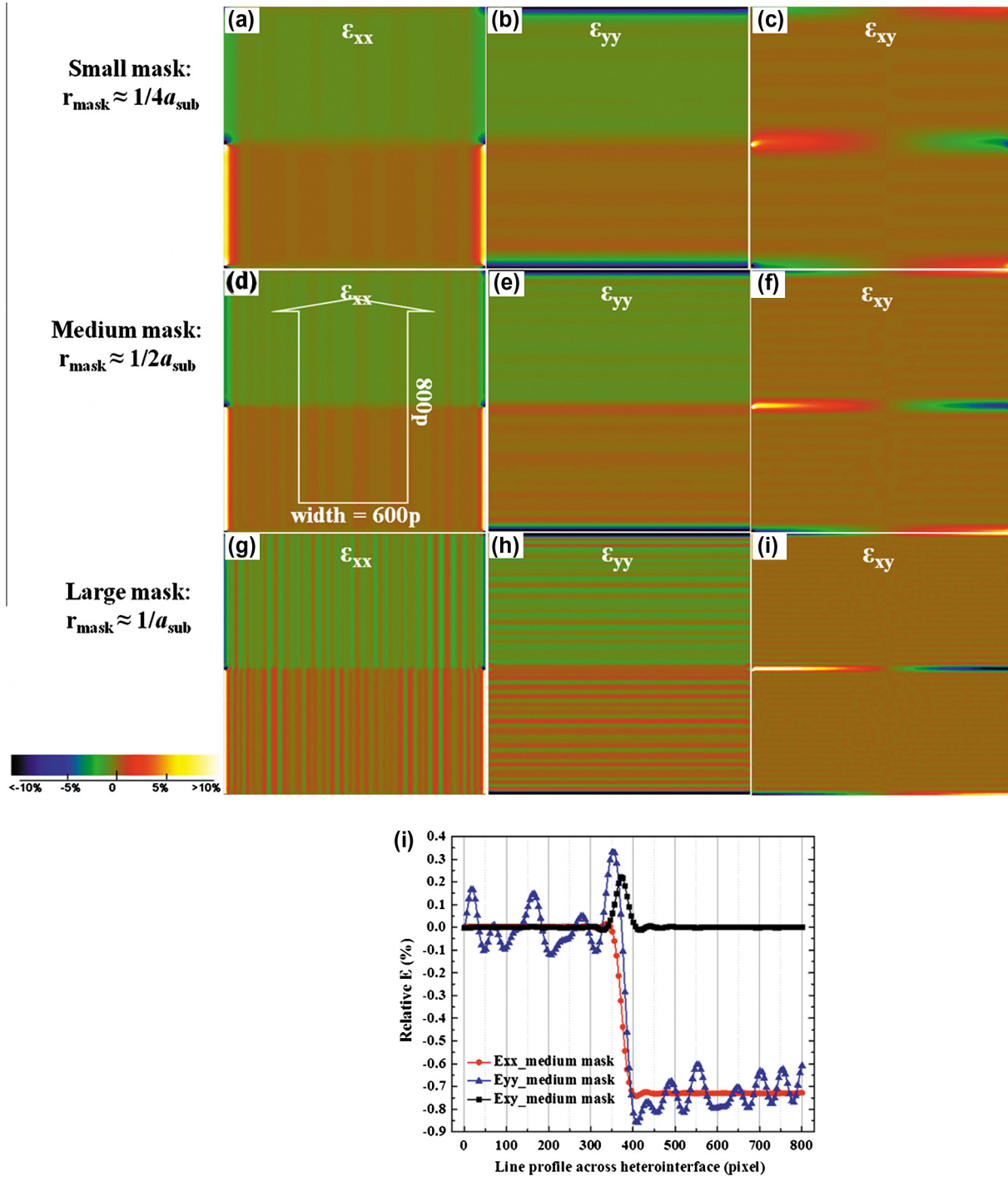


Fig. 2. (a–c) The GPA ϵ_{xx} (in-plane), ϵ_{yy} (out-of-plane) and ϵ_{xy} (shear strain) maps of computed image #1 using the small mask. GPA maps using (d)–(f) the medium mask and (g)–(i) the large mask. (i) The strain profiles of the medium mask set of relative ϵ_{xx} , ϵ_{yy} and ϵ_{xy} obtained across the interface. The profile size is noted in (d).

with almost fixed frequencies that are proportional to their mask sizes. Meanwhile, the size of the Fourier mask is reciprocal to the spatial resolution of the strain measurement in real space. That is to say, the small mask (in black) produces a smooth strain field but with a low spatial resolution, whereas a large mask (in blue) boosts spatial resolution (short length scale of the fluctuations) and is

accompanied by severe fluctuations, especially in ϵ_{yy} (e.g. Fig. 3e, where the strain variations are not averaged out by sampling). Moreover, in Fig. 3d and e, overshoots and undershoots were found (indicated by red¹ open arrows)

¹ For interpretation of color in Fig. 3, the reader is referred to the web version of this article.

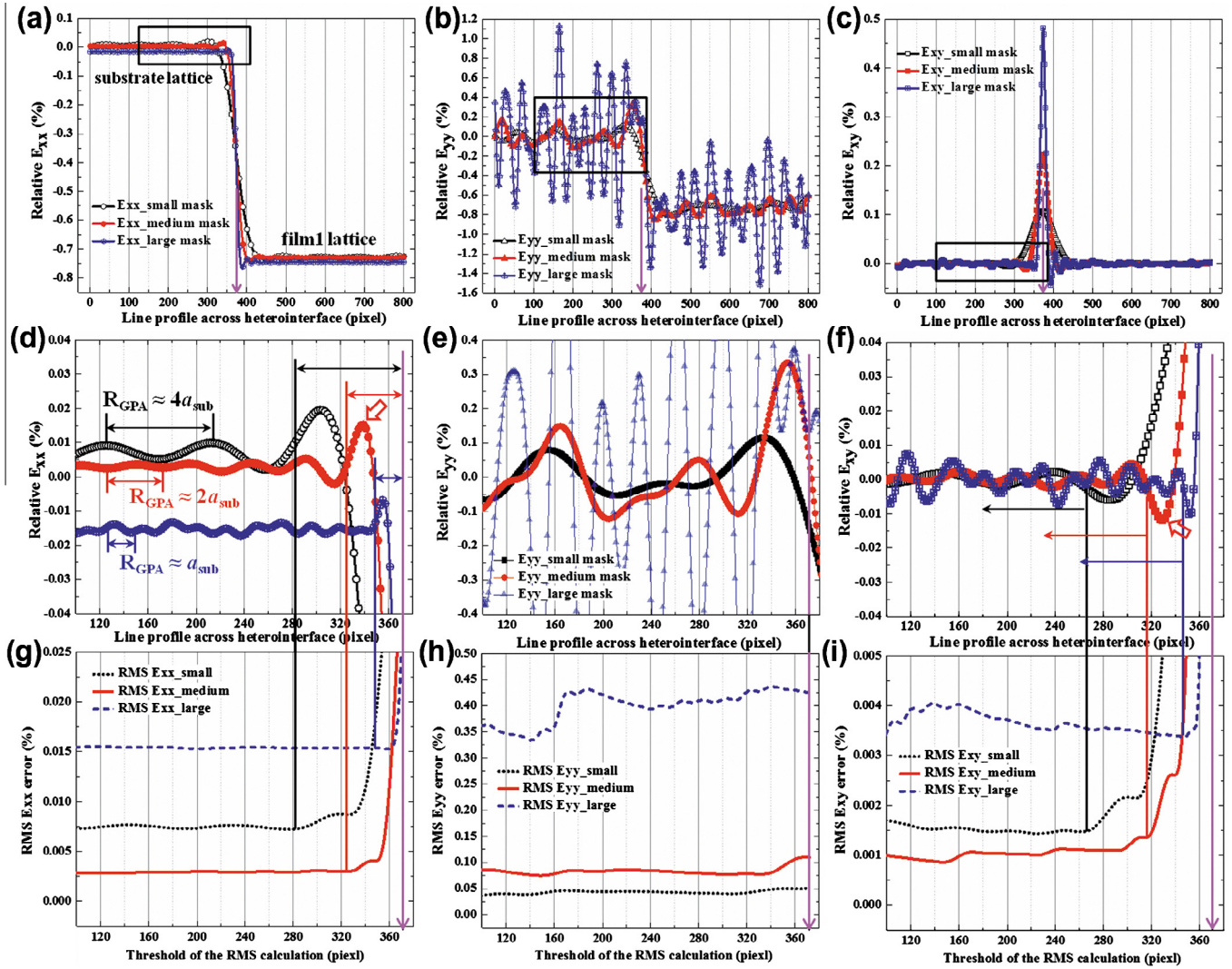


Fig. 3. The effects of the Fourier mask size on (a) ϵ_{xx} , (b) ϵ_{yy} and (c) ϵ_{xy} strain profiles across the interface. (d)–(f) Enlarged rectangular regions in the substrate in (a)–(c). The GPA spatial resolutions are noted in (d). (g)–(i) The root-mean-square (RMS) of the strains in (d)–(f) as a function of the profile threshold.

close to the interface (marked by purple arrows) irrespective of the mask size. Since there is no strain in the original computed image, these abnormal high/low strains are processing artifacts, which are sensitive to image quality and can introduce significant errors [34]. In addition, because they are located in the lattice layers next to the interface, it is necessary to quantitatively estimate their effective range from the interface. In Fig. 3g–i, the root-mean-square (RMS) variabilities of the strain profiles were calculated as a function of the data size, i.e. starting with the strain at pixel 120 to the interface, at pixel 121 to the interface, etc., to determine the theoretical GPA accuracy as well as to find out at which point the RMS errors start rising due to the overshoots/undershoots. The RMS errors in the constant strain regions are within 0.02%, 0.5% and 0.005% for the ϵ_{xx} , ϵ_{yy} and ϵ_{xy} profile strains, respectively, and about zero in the reference substrate region. Again, the values of the error depend partially on the profile direction (and the profile width, which is discussed later).

However, if a small Fourier mask (in black) is used, the GPA accuracy can be further improved to 0.01% of ϵ_{xx} , 0.05% of ϵ_{yy} and 0.002% of ϵ_{xy} that are all well within approximately 0.05% about zero. However, as marked in Fig. 3g and i, the drawback of this choice of mask is that the effects of the overshoots/undershoots on the strain error appear further from the interface than that of the other two larger masks: if one wants to reduce the abnormal strain near the interface by choosing a large mask (e.g. the blue line in Fig. 3g), the measurement of an accurate strain in the second unit-cell layer can be achieved but at the cost of the overall accuracy. This reflects the essential feature of local strain measurements, in which precision is a trade-off against spatial resolution. To balance the two, we suggest a medium mask size about twice the substrate lattice parameter in real space (the red plots in Fig. 3g–i), which gives an overall theoretical GPA accuracy within 0.1% and a measurable range of the third unit-cell layer from the interface.

In the above discussion, we found that not only the GPA processing parameters, e.g. the Fourier mask size, affect the strain accuracy, are important, but also how one surveys the strain maps and extracts the strain data matters. In addition to the directions of the strain profile, the profile width as shown in Fig. 3d and its effect on strain value were studied. Fig. 4a and b presents strain profiles of the ϵ_{xx} and ϵ_{yy} maps in Fig. 2d and e with different sampling widths. Compared to the known strain assigned in the lattice (marked with a dashed line as a visual guide), e.g. in the substrate region in Fig. 4a, the results given by profiles with small widths, e.g. 1 and 50 pixels, show deviations of about 0.1% and 0.05% that are obviously greater than those of other profiles (within 0.02%). The reason is that, as shown in Fig. 3d, the vertical ϵ_{xx} variations in Fig. 2d peak at a certain frequency of about $2a_{\text{sub}}$ (47 pixels). A profile width needs to be larger to avoid the error of insufficient sampling. As suggested in Fig. 4a, a profile width of 600 pixels is good for images 1024 pixels wide. In the case of the ϵ_{yy} strain profile, as shown in Fig. 4b, plots with different widths almost overlap, indicating that the strain distribution is independent of the sampling width due to the profile direction.

Factors associated with the features of source images and how they interplay with the inherent GPA processing and strain profile parameters were also examined. Image resolution, i.e. the numbers of pixels per lattice period, is considered as one of the main prerequisites for image-based lattice strain measurement [35,36]. To exclude possible influences in experimental imaging (e.g. recording media and TEM foil thickness [35]), computed images with low image resolutions of 512×512 and 256×256 pixels were used to test the effect of image resolution alone. The GPA strain profile results showed that as long as the width of the strain profiles is appropriate, the accuracy of the measured ϵ_{xx} and ϵ_{yy} strain barely changes with image size if at all (for details, see Supplementary Figs. S3 and S4). Thus, in the case of ideal image quality, the accuracy of the GPA strain profile is independent of the image resolution as long as the lattice distortion is larger than 1 pixel. Moreover, computed images with large lattice mismatches

between the substrate and the film (i.e. a sudden lattice change across the interface) have also been analyzed for the inherent GPA processing and strain profile accuracy. In Fig. 5, the GPA strain maps and profiles are presented for computed image #3, a model of the STO/MgO interface, with a large lattice mismatch (film #3 has a relative strain of -7.31% to the substrate). Comparing the ϵ_{xx} , ϵ_{yy} and ϵ_{xy} maps of image #3 in Fig. 5b–d with that of image #1 in Fig. 2d–f, there are six convergent regions, analogous to misfit dislocation cores, as the result of a relatively large lattice mismatch. These strain convergent regions correspond to (020) phase jumps caused by a translation of half an a -lattice fringe spacing in the top film lattice [12]. This is why the dislocation cores can be visualized in strain maps ϵ_{xx} and ϵ_{xy} , but not in ϵ_{yy} , which is related only to the 002 lattice displacement. Considering that these dislocation cores could affect the in-plane ϵ_{xx} strain measurement, two ways of strain profiling were performed, i.e. profiling at the position of the cores and between them on the computed image #2 with a minor mismatch. The results suggest that, with a reasonable profile width, profiling right at the cores gives a better accuracy, within 0.02% (for details, see Supplementary Fig. S5). Following this observation, strain profiling was performed at the dislocation cores in the strain maps ϵ_{xx} of image #3 (Fig. 5f), and the results show promising accuracy, despite a higher density of interfacial dislocation arrays. On the other hand, in Fig. 5g, the out-of-plane ϵ_{yy} profile of image #3 presents a greater (though not wider) undershoot in the STO film near the interface compared to that of the other two computed images. This processing artifact is likely due to the large lattice mismatch assigned in this interface (again, no lattice strain can be seen in the computed image). The interface model of image #3 also proves that GPA strain analysis is not limited to local lattices that have identical symmetry as the reference, though centrosymmetry is necessary [15].

In a short summary of Section 3.1, in the first part, four factors were studied using computer-generated images (the perfect crystal lattice), namely the Fourier mask size, strain profile width, source image resolution and misfit dislocations in the image, and the impacts of these factors on

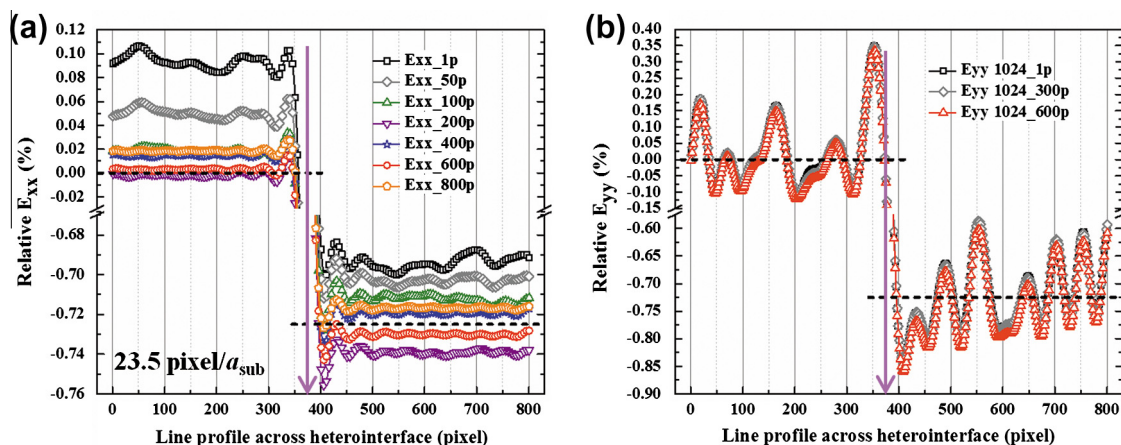


Fig. 4. The effects of profile width on (a) the ϵ_{xx} strain of Fig. 2d, and (b) the ϵ_{yy} strain of Fig. 2e across the interface.

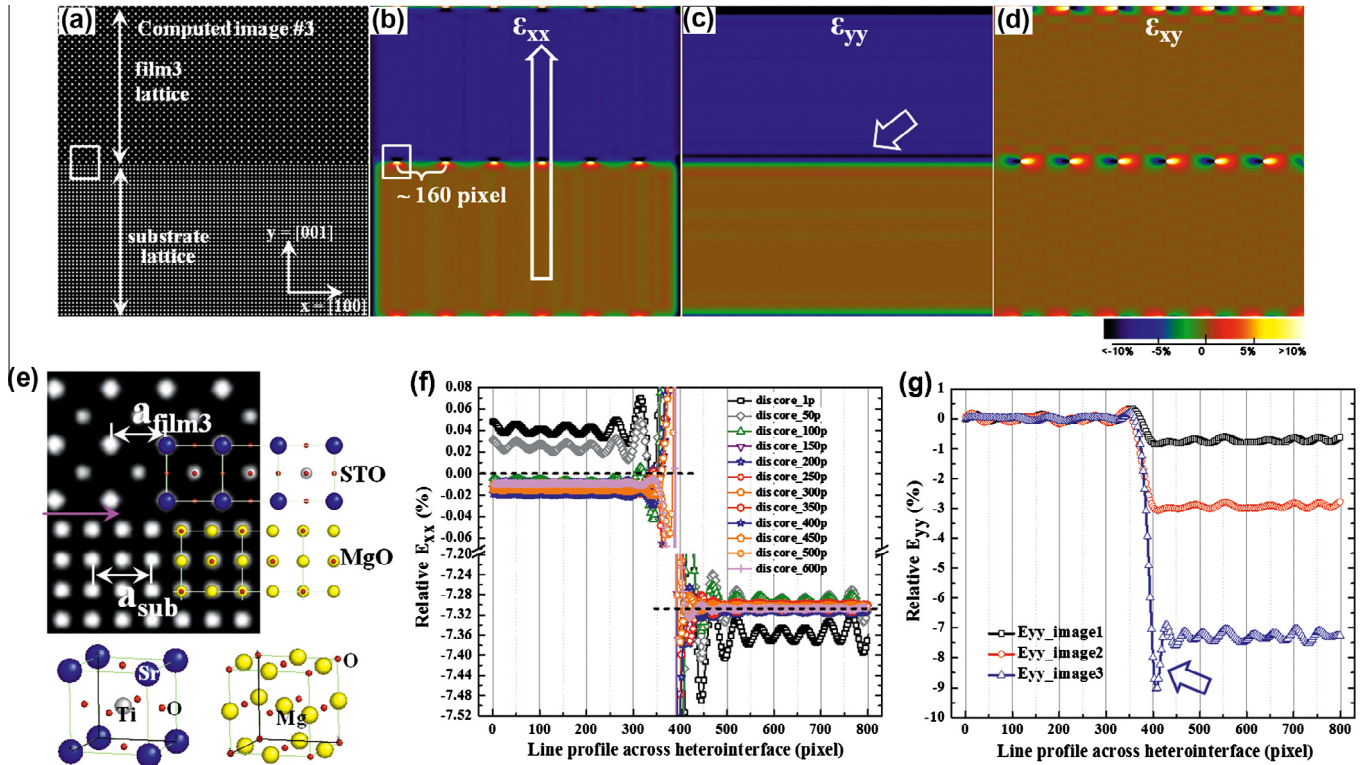


Fig. 5. (a) The computed image #3 of the heterogeneous structure of STO/MgO in $\langle 100 \rangle$. (b)–(d) The GPA ϵ_{xx} (in-plane), ϵ_{yy} (out-of-plane) and ϵ_{xy} (shear strain) maps of computed image #3 using a Fourier mask of $1/2a_{\text{sub}}$. (e) An enlarged image of the rigid interface between film #3 and the MgO substrate with corresponding atomic models. (f) The relative ϵ_{xx} strain profiles of (b) at the dislocation core as a function of profile width. (g) The relative ϵ_{yy} strain profiles of Figs. 2e, 6c and 5c across the interfaces. The severe undershoot in ϵ_{yy} of image #3 is marked by open arrows.

the inherent accuracy of GPA digital processing for quantifying lattice strain at thin-film heterogeneous interfaces were investigated. We found that mask size is crucial to the inherent deviation of GPA, which is a trade-off against the spatial resolution of the measurement; a medium mask size of $1/2a_{\text{ref}}$ is suggested as a compromise. In the profile direction across the heterointerface, this allows a theoretical GPA accuracy of 0.005% for ϵ_{xx} , 0.1% for ϵ_{yy} and 0.001% for ϵ_{xy} (Fig. 3g–i) and a spatial resolution of $2a_{\text{ref}}$ (about 47 pixels in Fig. 3). These GPA accuracies can be compromised if a small strain profile width is adopted. For an image of 1024×1024 pixels, an optimized profile width of 600 pixels is recommended (and at least 100 pixels are needed). No additional error was found when there are large lattice mismatch and associated misfit dislocations in an interface lattice image, as long as the strain profiles are properly applied, i.e. at the dislocation core and with a sufficient width. If the image contrast is uniformly sharp throughout the field of view, the resolution of the image (the number of pixels per lattice distance) is not a necessary requirement for high-accuracy GPA strain measurements.

3.2. GPA on Cs-corrected HAADF-STEM images of single crystals

GPA is capable of measuring the local lattice in Fourier space with high accuracy [29]. A precision of 3 pm was

achieved in a GPA strain field around dislocations using HRTEM [37]. Recent reports on STEM-image-based GPA give accuracies varying from 0.15% [38] to 0.40% [33]. In this section, Cs-corrected STEM images of single-crystalline substrate lattice far away from interfaces are used to determine the actual accuracy of STEM-GPA for interface strain quantification. Factors influencing the measurement precision are discussed.

STEM imaging is essentially different from TEM projections; the main artifact in STEM lattice images involves the distortions in the slow-scan direction, i.e. the so-called “fly-back” error [20,21]. This error can be clearly illustrated using GPA. In Fig. 6a, a high-resolution HAADF-STEM image was recorded by aligning the a -lattice axis of crystalline STO parallel to the fast scan direction (image resolution is 25.6 pixels per unitcell, and image size is 1024×1024 pixels). Using a Fourier mask of $1/2a_{\text{STO}}$, we obtained GPA strain maps of ϵ_{xx} (a -lattice displacement), ϵ_{yy} (c -lattice displacement) and ϵ_{xy} (shear strain) in Fig. 6b–d. Strain profiles were conducted along the out-of-plane c -direction to examine the lattice displacement across the ab -plane as in Fig. 2d–f, and the results of relative strains are presented in Fig. 6i. The RMS variation of the strain profiles was also calculated to estimate the error, which presents a very high RMS for ϵ_{yy} of 1.30% about zero. This dramatic deviation was caused by the “flyback” error in the slow-scan direction, appearing as the

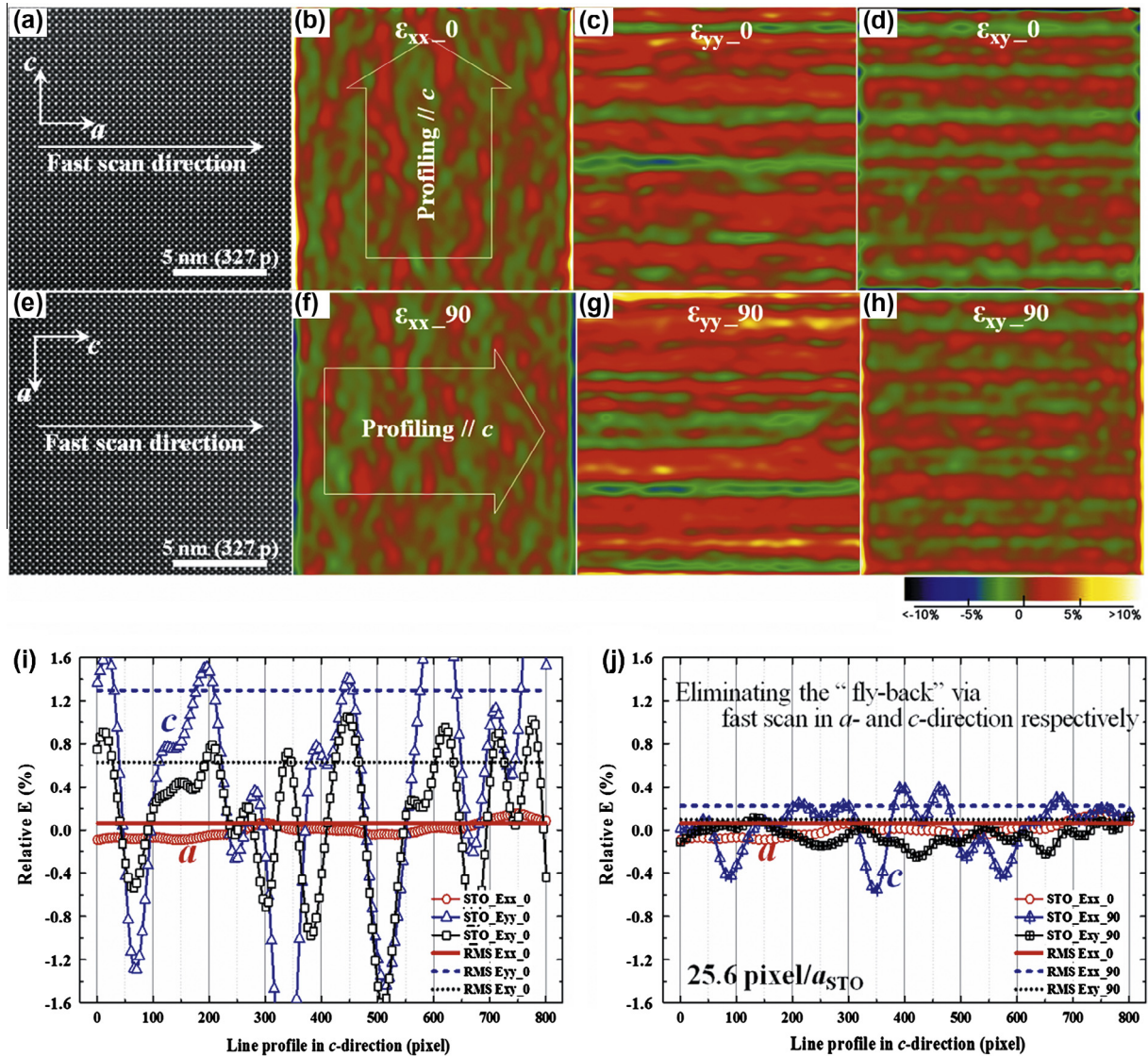


Fig. 6. (a) A representative Cs-corrected STEM image of the STO substrate with *a*-lattice direction parallel to the fast scan direction in $\langle 100 \rangle$. (b)–(d) The GPA ϵ_{xx} , ϵ_{yy} and ϵ_{xy} maps of (a) using a Fourier mask of $1/2a_{\text{sub}}$. (e) The STO substrate STEM image obtained with *c*-lattice direction aligning to the fast-scan direction. (f)–(h) The GPA ϵ_{xx} , ϵ_{yy} and ϵ_{xy} maps of (e). (i) The relative strain profiles and RMS value of the strain maps of image (a). (j) The relative strain profiles and RMS value of the ϵ_{xx} in (b), the ϵ_{xx} in (f) and the ϵ_{xy} in (h). Note that both of the ϵ_{xx} in (b) and (f) are along the fast-scan direction, and lead to small RMSs in *a*- and *c*-displacements.

horizontal stripes in the ϵ_{yy} strain map. The imaging direction in Fig. 6a introduces severe artifacts in the measurement of the *c*-lattice displacement.

Various methodologies, such as digital correction algorithms [20], have been proposed to correct this systematic distortion. Considering that digital processing might have difficulty dealing with interfacial lattice jumps at interface defects, and that the “flyback” error is sensitive to STEM scan direction [21], we adopted a simple solution of recording the same image twice with a 90° scan rotation, aligning the fast scan direction parallel to the *a*- and *c*-direction, respectively. The second set of STEM images and GPA strain maps is shown in Fig. 6e–h, with the ϵ_{xx} noted as ϵ_{xx_90} to distinguish it from the ϵ_{xx_0} in Fig. 6b. Note that since ϵ_{xx_90} is profiled across the *ab*-plane (where the hetero-

ointerface lies in subsequent interface applications), it represents *c*-lattice displacement. Combining the two strain maps, which are both in the fast-scan direction, we obtained quantitative relative *a*-lattice and *c*-lattice strains in Fig. 6j. After eliminating the distortion in the slow-scan direction, in Fig. 6j a much lower RMS error of 0.22% (about zero) for the *c*-lattice strain is achieved. In the case of the lattice shear distortion ϵ_{xy} , based on its definition of $\epsilon_{xy} = \frac{1}{2}(\partial u_x(r)/\partial y + \partial u_y(r)/\partial x)$, the result has to be partially dependent on the slow-scan distortion in either imaging orientation. One way to obtain full accuracy for ϵ_{xy} could be to take one term from each image (the $\partial u_x(r)/\partial y$ term from the *x* fast scan image, and the $\partial u_y(r)/\partial x$ term from the *y* fast scan image), under the assumption that the two images are collected from identical sample regions. Since the two

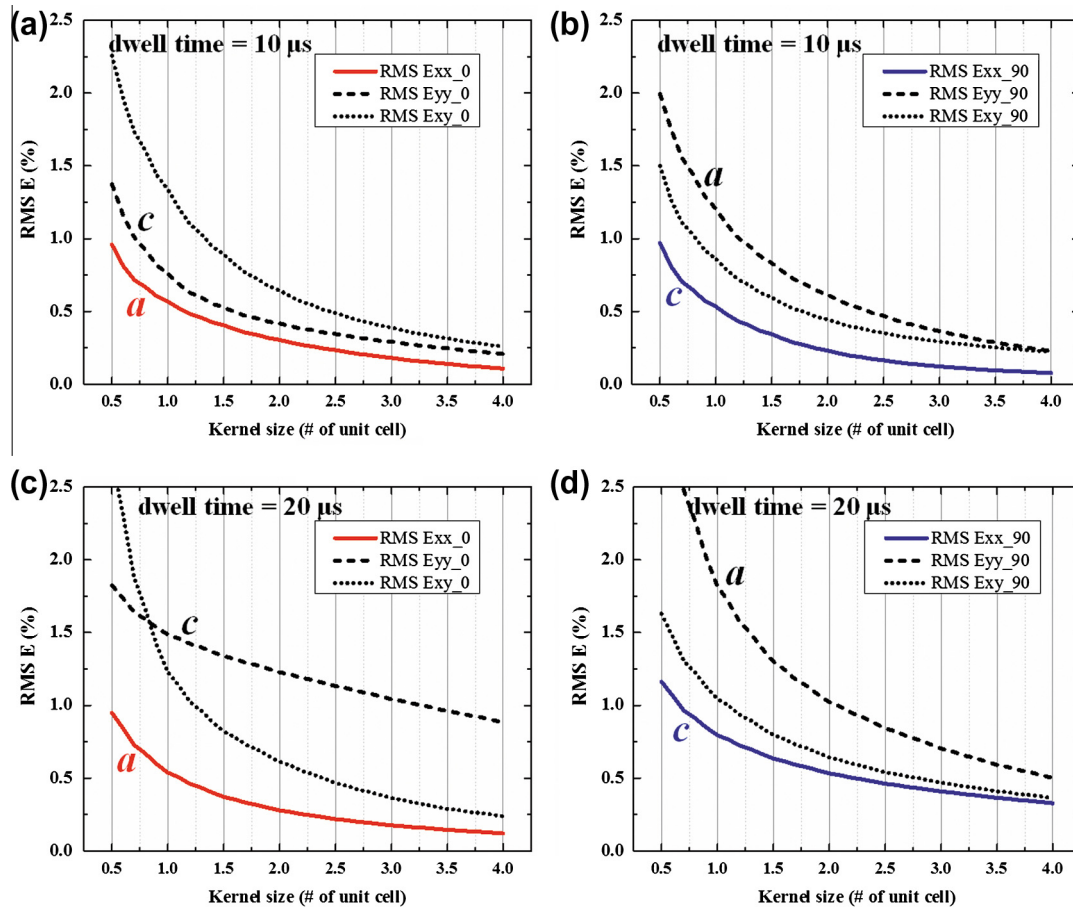


Fig. 7. The experimental STEM-GPA accuracy assessed by real space strain measurements. The RMS strains as a function of kernel size in a STEM image of the STO substrate with (a) the a -lattice direction and (b) the c -lattice direction parallel to the fast scan direction with the default dwell time of 10 μs . The RMS strains in a STO STEM image obtained with a dwell time of 20 μs with the fast-scan direction in (c) the a -lattice direction and (d) the c -lattice direction.

images were recorded sequentially rather than simultaneously, there is no guarantee for absolutely perfect alignment between the two. An alternative solution is to use the ε_{xy} obtained after the 90° rotation ($\varepsilon_{xy,90}$ in Fig. 8j), because some of the slow-scan distortion is canceled out along this profile direction. We repeated the above GPA profiles for several pairs of STEM images of the single-crystal STO taken under similar conditions, and found that the GPA accuracy using STEM images, as shown in Fig. 6j for example, is approximately within 0.25%.

To further verify the above error estimation, a real-space strain measurement method was also used to quantify the RMS strains in the above single-crystal STEM images as a function of kernel size (the Fourier mask in real space). Fig. 7a and b are a set of representative results obtained from a pair of STEM images before and after a 90° rotation. In both plots, the strains mapped along the fast-scan direction are shown by solid lines; dashed lines indicate the strains along the slow-scan direction. For a kernel size of two unit cells, the RMS of the a -displacement in Fig. 9a and of the c -displacement in Fig. 9b are both well within 0.25%, agreeing with the above GPA assessment (Fig. 6j). The strain accuracy increases with kernel size,

consistent with the effect of changing the Fourier mask size on GPA precision, as discussed above. Again, a larger kernel size (or a smaller Fourier mask) leads to higher accuracy, but at the cost of spatial resolution. It is also noted that the RMS strains become almost constant as the kernel size increases. We found that how fast and how well the measured RMS strain approaches zero depends greatly on the direction of the residual drift of the TEM sample. Even though we waited until there was no noticeable drift in the field of view, there is always some drift during the time required to obtain a STEM image. For example, in the case of Fig. 6, the residual drift is mainly in the c -lattice direction, because the RMS strain of c -displacement is obviously larger when it is along the slow-scan direction in Fig. 6i than along the fast-scan direction in Fig. 6j. An opposite example is the measurement in Fig. 7a and b, where the direction of the residual sample drift (defined as drift that is not a linear function of time) is close to the a -lattice axis. When the sample drift direction aligns with the slow-scan direction, a large error appears. This could harm the accuracy even after using the 90° rotation recording. As shown in Fig. 7c and d, this set of RMS strains was measured from the STEM images obtained

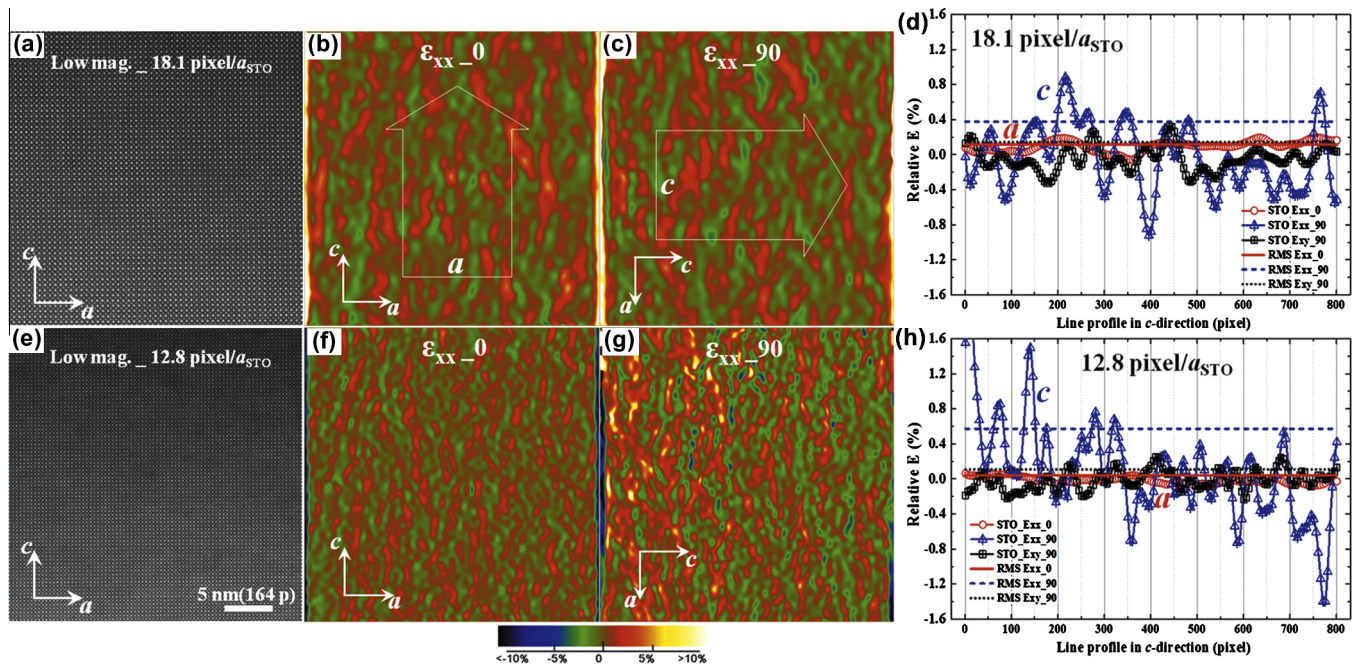


Fig. 8. The effects of STEM image resolution on the actual GPA accuracy. (a) The STEM images with the same view field as Fig. 6a with a lower resolution of 18.1 pixel/ a_{STO} . (b)–(d) The a -lattice strain map, the c -lattice strain map and relative profiles in the out-of-plane directions. (e) The STO STEM images of 12.8 pixel/ a_{STO} and (f)–(h) the corresponding GPA strain profile results. Note that the RMS of the c -displacement increases as the image magnification decreases.

using a longer dwell time of 20 μs . After aligning the c -lattice axis to the fast-scan direction, the RMS strain of c -displacement decreases compared to that in Fig. 7c; the residual drift limits the accuracy to 0.35% regardless of the kernel size. Thus, the residual sample drift in the STEM fast-scan direction (related to the dwell time) determines the scanning artifacts in this image-based lattice strain quantification.

As recalled from Section 3.1, it is found that the resolution of the computed images does not affect the GPA strain accuracy; however, this is not the case for the experimental images. In Fig. 8a–h, the GPA strain mappings and profiles of two STEM images with lower resolutions of 18.1 pixel/ a_{STO} and 12.8 pixel/ a_{STO} , respectively, are presented. The source STEM images were recorded around the same field of view as Fig. 6a and e. 90° rotation recording was also applied to minimize the slow-scan distortion. Comparing the strain profiles in Fig. 8d and h with that in Fig. 6j, the RMS of the a -lattice strain remains in the same range, whereas the RMS of the c -displacement increases as the image magnification decreases; it reaches 0.57% in Fig. 8h, more than double the 0.22% error obtained in Fig. 6j. From a survey of the intensity line profile, we found this degradation of accuracy to be closely related to the local image focus. The optimum focus of HAADF-STEM is a balance between a sharp probe and minimizing probe tails; in practice, the focus was chosen to maximize contrast [17]. For the high-magnification STEM images (e.g. Fig. 6a and e), we tested STO regions with different sample thicknesses, estimated from 16 to 48 nm by EELS, and a strain

RMS error within 0.25% can be achieved when the contrast is maximized by choosing a suitable focus. However, in the case of low-magnification images, a uniform focus is difficult to obtain over a large field of view. Since the image-based lattice strain measurement depends on the positions of atomic columns (or contrast maxima), defocus caused by sample thickness variation introduces strain deviations at the edge of the image in Fig. 8e and increases the overall RMS error. For detailed comparisons of focused and unfocused local regions in low-magnification STEM images, see Supplementary Fig. S6. In fact, thickness and focus variations have been previously identified as major concerns in HRTEM-based GPA [16,34]. As for STEM, the maximum contrast criterion provides optimum focus at the region of interest; however, some parts of the sample can be out of focus when the view field is large and the sample thickness is uneven or the foil is bending. In Fig. 8, the reason why the RMS of the a -displacement does not drop as much as the c -strain is because the thickness variation in the cross-sectional TEM sample is mainly along the c direction (the out-of-plane film growth direction), whereas the sample thickness in the a direction is relatively uniform in the field of view. The RMS of the a -displacement could also be affected (see Supplementary Fig. S8). Although STEM is more tolerant of contrast reversals from thickness/focus change, high-quality (not necessarily high-resolution) lattice images with uniformly sharp contrast are still a prerequisite for accurate strain determination.

Based on the GPA strain profiles on Cs-corrected HAADF-STEM images of the single-crystal STO substrate

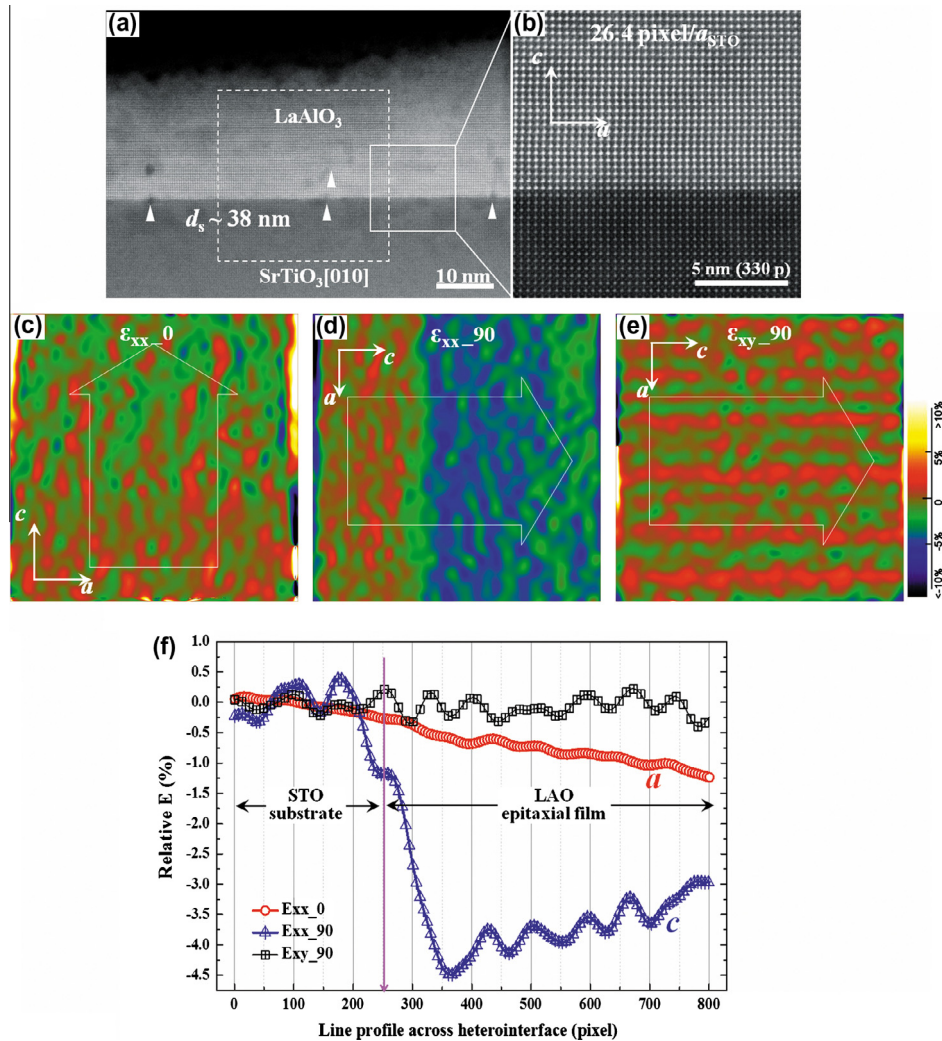


Fig. 9. (a) A representative cross-sectional STEM overview of the 50 nm thick LAO epitaxial thin film on STO (100) in the $\langle 100 \rangle$ zone axis. (b) A high-resolution STEM image of the LAO/STO interface without dislocations. (c)–(e) The GPA a -lattice strain, the c -lattice strain and shear strain maps of (b). (f) The relative strain profiles of (c)–(e) across the LAO/STO heterointerface.

lattice, an actual accuracy within 0.25% was determined for both a - and c -lattice displacements. Considering that STO bulk lattice parameter is about 0.3905 nm, the STEM-GPA precision is ~ 1 pm. This high accuracy can be experimentally achieved after eliminating the STEM slow-scan distortion by the 90° rotation recording method. Two factors were found to play major roles in influencing the STEM-GPA precision: (i) the image focus and its degree of uniformity across the source STEM image; and (ii) the residual sample drift in the STEM fast-scan direction. Unlike computer-generated images, where the contrast is ideal, some regions of the experimental lattice images could be out of focus due to variation in the sample thickness, especially when the field of view becomes large. Unfocused lattice images introduce uncertainties in atom column position, and thus errors in determining lattice displacement. If an optimum focus is satisfied across a region of interest, the strain measurement accuracy then depends mainly on the residual drift in the fast-scan direction. A simple way of

checking whether the sample has reached thermal equilibrium and whether the dwell time is appropriate in terms of balancing noise and drift is to identify if the image is free of apparent strain in the crystalline lattice away from defects.

3.3. GPA on Cs-corrected HAADF-STEM images of heterogeneous interfaces

In this part, we applied the above STEM-GPA strain profile method to quantify the lattice strain of two representative heterostructures, i.e. $\text{LaAlO}_3/\text{SrTiO}_3$ and $\text{SrTiO}_3/\text{MgO}$ interfaces. The overall film crystallographic properties were first characterized by XRD θ – 2θ scans (for XRD plots, see [Supplementary Fig. S7](#)). Based on the sharp primary (00 l) peaks, the out-of-plane lattice parameters (c) were calculated to be 0.377 and 0.390 nm for the LAO and the STO thin films, respectively, calibrated based on the single-crystal substrates. Assuming that the film lat-

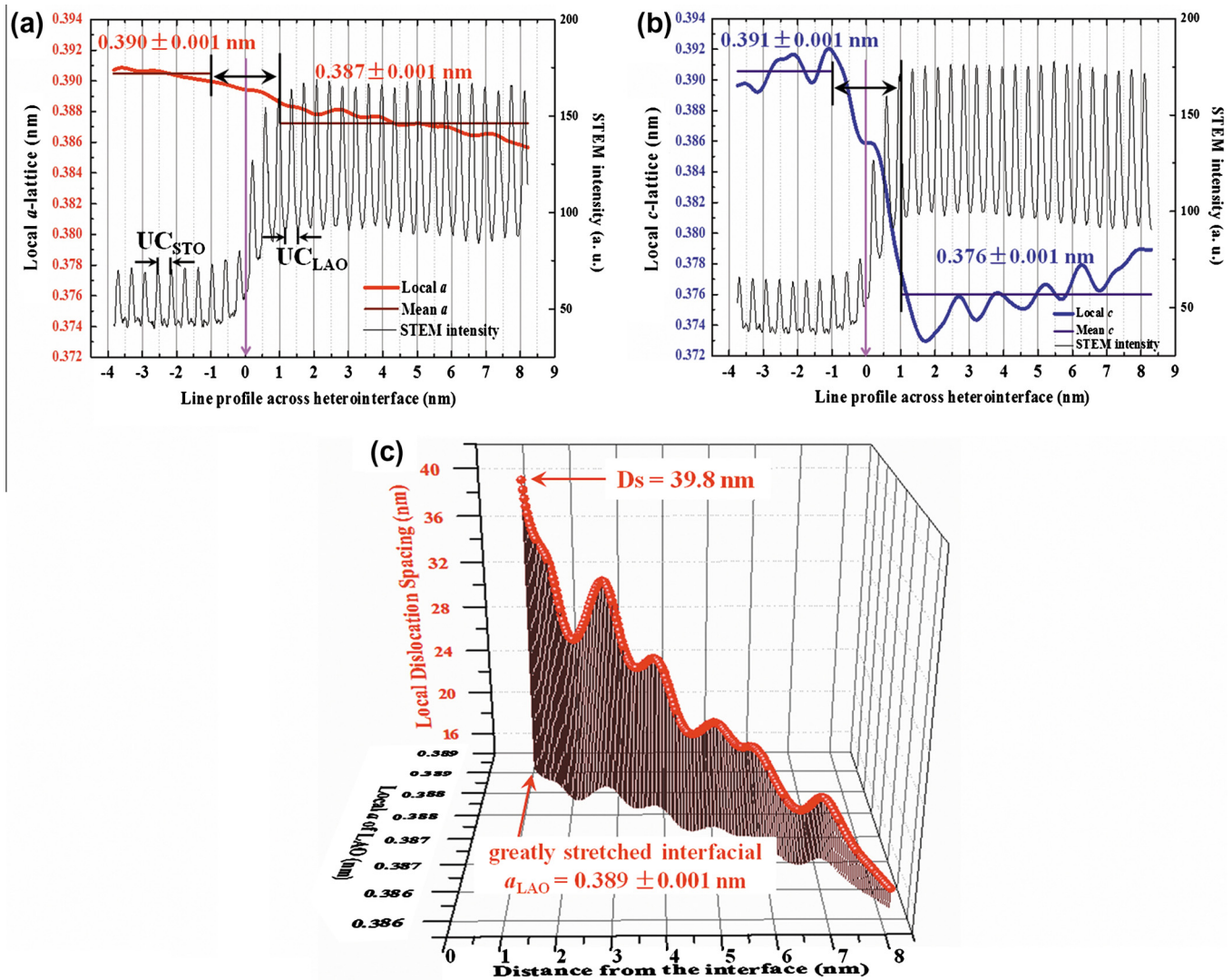


Fig. 10. The local and mean (a) a -lattice parameters and (b) c -lattice parameters based on the strain profiles in Fig. 9f as a function of the distance to the interface in nanometers. The intensity profile across the LAO/STO interface was superimposed to illustrate the position of the local lattice. (c) The local dislocation spacing D_s as a function of the actual LAO film a -lattice from the interface.

tice parameter a is equal to c (cubic structure), the interface misfit rate ($f = 2(a_s - a_f)/(a_s + a_f)$) and corresponding misfit dislocation spacing ($D_s = (a_f \times a_s)/2(a_f - a_s)$) were calculated to be 3.65% and 5.26 nm for the $\text{LaAlO}_3/\text{SrTiO}_3$, and 7.64% and 2.65 nm for the $\text{SrTiO}_3/\text{MgO}$ heterogeneous interface, respectively.

Fig. 9a is a representative cross-sectional STEM overview of the 50 nm thick LAO epitaxial thin film on STO (100) substrate acquired in the $\langle 100 \rangle$ zone axis. Three misfit dislocations, marked by arrows, were found at the heterogeneous interface of LAO/STO with a dislocation spacing (D_s) of about 38 nm. This is much larger than the dislocation spacing of 5.26 nm estimated by XRD, which suggests a much smaller in-plane lattice mismatch between the LAO film and the STO substrate. To determine the actual lattice displacement of this strained heterogeneous interface, we applied the GPA strain profile to a high-resolution STEM image of the interface (Fig. 9b). The GPA strain maps of

$\varepsilon_{xx} \sim 0$ (a -lattice displacement), $\varepsilon_{xx} \sim 90$ (c -lattice displacement) and $\varepsilon_{xy} \sim 90$ (shear strain) were generated using a Fourier mask of $1/2a_{\text{STO}}$ and are presented in Fig. 9c–e. Using a profile across the heterointerface with a width of 600 pixels (~ 12 nm) suggested by the findings in Section 3.1, the relative lattice strains with respect to the STO substrate are plotted in Fig. 9f. It is interesting to see that the relative a -lattice strain, instead of showing an instantaneous jump in the c -lattice strain, shows a gradual decline across the heterointerface. The shear strain ε_{xy} oscillates about zero, indicating there is little out-of-plane rotation in both the film and substrate lattices.

To understand the above strain profiles, we replotted the relative a -lattice strain into the local a -lattice parameters as a function of the distance to the interface in nanometers in Fig. 10a. The intensity profile across the LAO/STO interface was superimposed to illustrate the correlations between the value of a and the position of the local lattice.

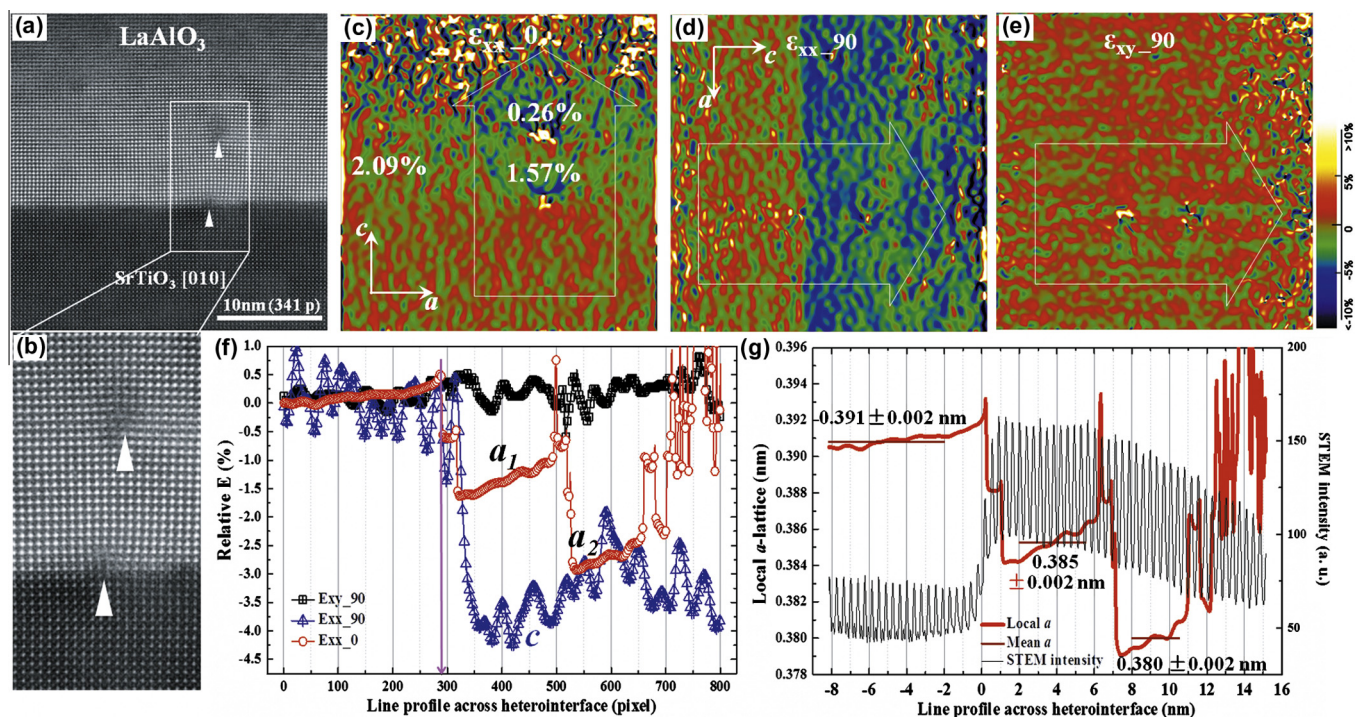


Fig. 11. The secondary strain relaxation observed in LAO/STO GPA strain profiles on STEM images with misfit dislocations. (a) A STEM image of the LAO/STO interface with dislocations. The image is from the field of view marked by the dashed square in Fig. 9a. (b) Enlarged view of the dislocations. (c)–(e) The GPA a -lattice strain, the c -lattice strain and shear strain maps of (a). (f) The relative strain profiles of (c)–(e) across the LAO/STO heterointerface. (g) The local and mean a -lattice parameters of (c) as a function of the distance to the interface in nanometers.

It is clear that near the interface of LAO/STO, indicated by a vertical arrow, the in-plane a -lattice relaxes in a gradual manner. To assess the a -lattice spacing near the heterointerface, considering that the overshoots/undershoots at interfaces are inherent GPA digital processing artifacts (e.g. Fig. 3d), the lattice regions within 1 nm to the interface ($\sim 2a_{\text{STO}}$, i.e. the spatial resolution of GPA) were excluded from the calculation. Then, by averaging in selected regions in Fig. 10a, the a of the STO substrate was calculated to be 0.390 ± 0.001 nm and the a of the initial 8 nm thick LAO film is 0.387 ± 0.001 nm. The measured a of the substrate fits well with the XRD results and the STO bulk parameter; however, the a -value of our LAO film is considerably larger than its bulk counterpart ($a_{\text{LAO}}^{\text{bulk}} = 0.379$ nm). This strongly suggests that the initial film in-plane lattice is in lateral tension, 2.09% larger than the LAO bulk parameter. Meanwhile, the results of the c -lattice spacing near the interface in Fig. 10b show that the initial film out-of-plane lattice (0.376 ± 0.001 nm) is slightly smaller than the bulk parameter, consistent with the out-of-plane compression introduced by lateral tension in the initial LAO film. When we repeated the above measurement in other STEM images of the LAO/STO interface (without misfit dislocations), even for the images with lower magnification, similar results on the local lattice parameters were obtained (see Supplementary Fig. S8). The STEM-GPA strain profile proved to be reproducible and reliable.

Based on the local lattice parameters measured above, some important interface properties can be revealed. For example, the tetragonality of perovskite oxides, defined as c/a , is considered to be closely related to ferroelectric polarization [39]. Here, it yields a tetragonality value of $c/a \approx 1.03$ in the initial 8 nm thick LAO film. In Fig. 10c, we calculated the local dislocation spacing D_s as a function of the actual LAO film a -lattice at the interface. A D_s of about 40 nm was found as a result of the severely stretched interfacial film lattice; this dislocation spacing is consistent with the spacing of 38 nm observed directly at the interface (Fig. 9a). Furthermore, it is noted that besides the misfit dislocations at the interface, additional dislocations are observed in the LAO thin film (Fig. 9a). We applied the GPA strain profile to a STEM image including these misfit dislocations (the field of view is marked by a dashed square), and the results are presented in Fig. 11. It is interesting to see that the LAO film a -lattice strain no longer relaxes gradually, but instead jumps instantaneously across the misfit dislocations. Although this image is less uniform and gives a higher error in local lattice quantification, as shown in Fig. 11g the first 6 nm LAO layer has an a of 0.385 ± 0.002 nm, and it is relaxed to 0.380 ± 0.002 nm by generating the second misfit dislocation in the LAO film about 6 nm away from the interface. Combining the results in Figs. 9 and 11, a more complete in-plane film strain distribution can be obtained. Based on the local a -lattice, in strain map Fig. 11b, the absolute LAO film lateral strains

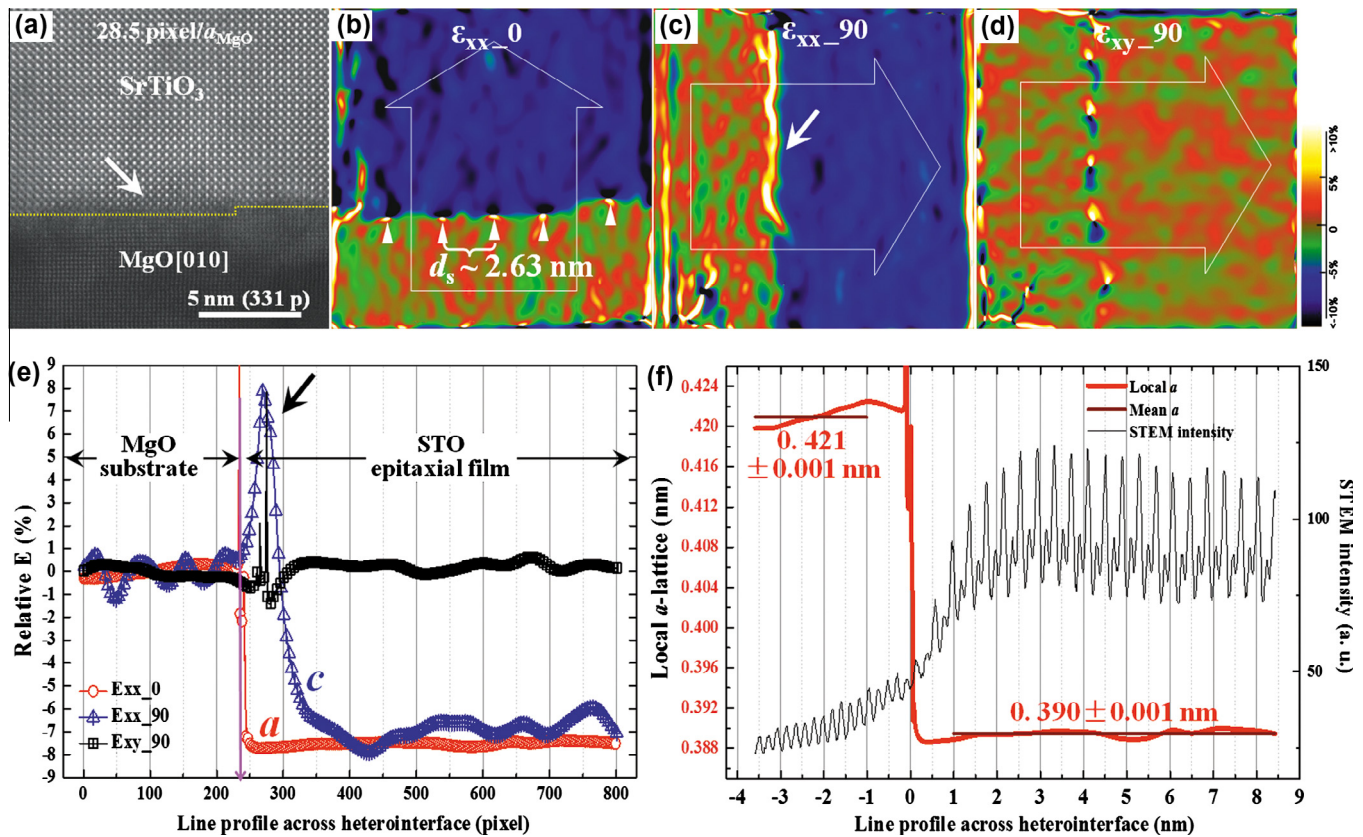


Fig. 12. (a) A representative high-resolution STEM image of the STO/MgO interface in $\langle 100 \rangle$. (b)–(d) The GPA a -lattice strain, the c -lattice strain and shear strain maps of (a). (e) The relative strain profiles of (b)–(d) across the STO/MgO heterointerface. (f) The local and mean a -lattice parameters of (c) as a function of the distance to the interface in nanometers. The interface steps and interdiffusion are marked by arrows.

are 2.09% in between dislocations, 1.57% in the initial 6 nm LAO layers after relaxation by dislocations at the interface, and 0.26% after the secondary relaxation. Then, the LAO film is almost fully relaxed to the bulk parameter. These experimental observations and quantifications provide the actual lattice strain distribution at the LAO/STO heterointerface.

In the case of the STO/MgO heterointerface, the GPA strain profile results are in good agreement with the XRD prediction. A representative high-resolution STEM image of the 40 nm thick STO/MgO heterostructures in the $\langle 100 \rangle$ projection is presented in Fig. 12a. The GPA strain maps were obtained in Fig. 12b–d using a Fourier mask of $1/2a_{\text{MgO}}$. In the strain map ϵ_{xx} , the convergent region of strain around the dislocation core clearly demonstrates that the dislocation spacing is about 2.63 nm, consistent with the value estimated from the XRD out-of-plane measurement (Fig. S7b). Strain profiles were also measured (as marked by the open arrows) and are presented in Fig. 12e. Here, the STO film in-plane a -displacement drops immediately once across the interface and remains almost constant in the film. In Fig. 12f, the mean a of the initial 8.5 nm thick film is nearly the same as the bulk value, indicating a very small residual lateral strain in the film. The out-of-plane c -displacement shares a similar trend as that of the a -displacement. In Fig. 12e, the dra-

matic overshoot near the interface reflects the actual interface imperfections of MgO substrate surface terraces and interdiffusion (marked by arrows). (For detailed discussions on the misfit dislocations and associated interface defects of the STO/MgO heterostructures, see Ref. [40].) Thus, based on the GPA strain profiles above, we found that at the STO/MgO heterointerface the lateral film strain has been relaxed by generating a high density of misfit dislocations, presenting a prototype domain-matching epitaxy [41].

3.4. Discussions on the limitations of GPA

In addition to the trade-off between the spatial resolution and the precision, which is a feature of local measurement averaging, there are some essential limitations to GPA strain profiles. One of these limitations is due to the nature of GPA, which is basically a comparison process. Since it measures a relative lattice parameter with respect to a selected reference, the observed structure needs to be fairly similar to the reference.

An example to demonstrate the limitation of GPA is shown in our attempt to quantify the heterointerface of FeSe_{0.5}Te_{0.5}(FST)/STO in Fig. 13. In this case, as shown in the enlarged $\langle 100 \rangle$ projection (Fig. 13b, in which the unit cell of the film is marked by a rectangle), the

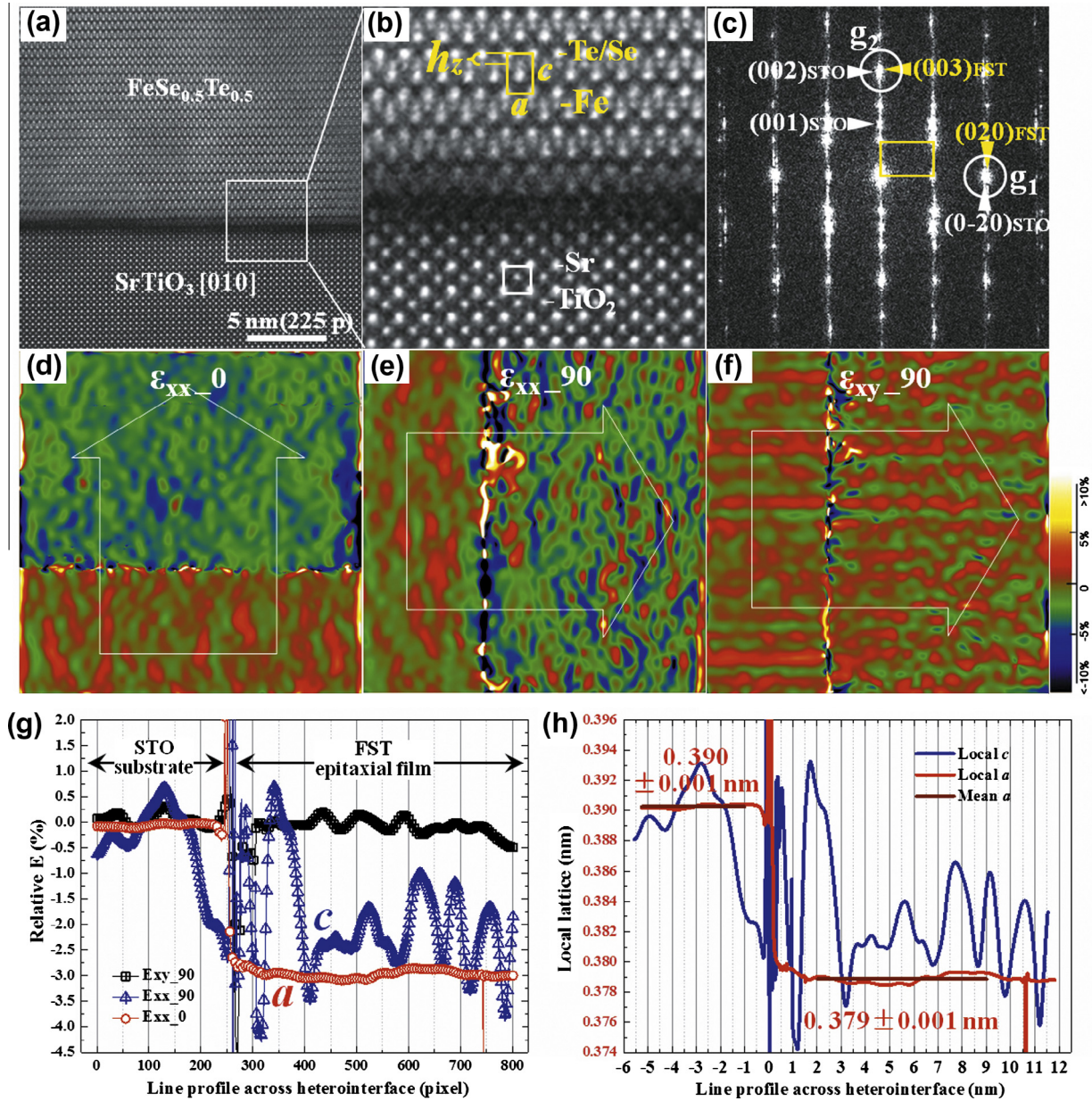


Fig. 13. (a) A representative high-resolution STEM image of the FeSe_{0.5}Te_{0.5}/STO interface in (100). (b) Enlarged view of the FeSe_{0.5}Te_{0.5} and STO lattice with the unit cell marked. (c) The power spectrum of (a). The Fourier vectors $g_1 = [020]_{STO}^*$ and $g_2 = [002]_{STO}^*$ were selected for GPA. (d)–(f) The GPA *a*-lattice strain, the *c*-lattice strain and shear strain maps of (a) using a mask size of $1/2a_{STO}$. (g) The relative strain profiles of (d)–(f) across the FST/STO heterointerface. (h) The local and mean *a*-lattice and *c*-lattice parameters as a function of the distance to the interface in nanometers.

FeSe_{0.5}Te_{0.5} film has a *P4/nmm* tetragonal structure (space group #129 [42]). The lattice parameters of the FeSe_{0.5}Te_{0.5} film were found to be $a_{FST} = 0.380$ nm, $c_{FST} = 0.586$ nm from electron diffraction [43]. Comparing with the perovskite STO substrate (bulk $a_{STO} = c_{STO} = 0.390$ nm), c_{FST} is about 1.5 times larger than c_{STO} . From the power spectrum in Fig. 13c, the choice of Fourier vectors $g_1 = [020]_{STO}^*$ and $g_2 = [002]_{STO}^*$ (close to the (003)_{FST} reflection) can still be satisfied using a mask size of $1/2a_{STO}$. However, as shown in the relative strains obtained from the GPA maps (Fig. 13d–f), in Fig. 13g the out-of-plane *c*-displacement presents severe oscillations. As we transfer the relative

strains into local lattice parameters in Fig. 13f, the mean *a* of the initial 9 nm thick FST film is 0.379 ± 0.001 nm, consistent with the SAED results, whereas the local *c*-parameter in the film is too random to enable further reasonable calculation. If the difference between the structure being measured and the reference is larger or smaller than half the reference lattice parameter (i.e. half of the periodicity π), there will be phase jumps (discontinuities) in the phase image due to a phase restriction of $-\pi$ to π in the GPA algorithm [12]. In practice, this leads to noisy strain maps with overshoots/undershoots in every unit cell. Moreover, due to the periodic nature of the GPA method-

ology, lattice distortion within a single unit cell cannot be quantified. The chalcogen height (h_z), defined as the vertical distance of the Se/Te plane above the Fe plane (marked in Fig. 13b), is one of the most important lattice parameters for Fe-based superconductors [44]. It would be a valuable complement to diffraction methods if h_z could be quantified using the image-based measurement at a higher spatial resolution. However, as shown in the GPA strain profile in Fig. 13, determining the dependence of h_z on the atomic coordinates within a unit cell is beyond the capability of GPA.

4. Conclusions

In this work, the accuracy and reliability of using GPA to measure interface lattice strain were systematically investigated, using both computer-generated and experimental Cs-corrected HAADF-STEM images of single-crystalline lattices and heterogeneous interfaces with different lattice misfits. The conclusions and guidelines identified are as follows:

1. The inherent accuracy and spatial resolution of GPA depends greatly on the mask size, the strain profile direction and width, but not on the image resolution (with ideal contrast). A medium mask size of $1/2a_{\text{ref}}$ is suggested to balance precision and resolution for local measurement. Strain profile widths approaching 1 pixel should be avoided; sufficient sampling is necessary to obtain representative strain profiles across a heterointerface.
2. An experimental strain accuracy within 0.25% (lattice precision of 1 pm) was achieved in STEM-image-based GPA after eliminating the STEM slow-scan distortion by recording two images with scan directions 90° apart. A good consistency check is to identify if the image in the fast-scan direction is free of apparent strain in the crystalline lattice away from defects.
3. The size of the field of view for GPA strain quantification with high precision is limited by the degree of uniformity in the STEM image contrast of the sample. Special care needs to be taken with TEM foil preparation to achieve a uniform sample thickness.
4. The STEM-GPA strain profile has been proven effective in quantifying the lattice displacement fields of the partially strained LAO/STO heterointerface with a secondary relaxation mechanism, and of the fully relaxed STO/MgO heterostructures with a high density of interface misfit dislocations. Repeating the GPA processing for STEM images obtained under different imaging conditions (focus, view fields, magnification, etc.) provides a better understanding of the overall strain field.
5. STEM images, especially the Cs-corrected HAADF images, are reliable sources for image-based lattice strain quantification. Although thickness/focus variations can affect the image contrast from one region to another (and hence the overall accuracy), they are not very detrimental to the GPA strain profile. However, there are some fundamental limitations when applying GPA to a structure whose lattice parameter is very different from that of the available reference lattice.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.actamat.2013.06.006>.

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